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**Research Article/ Review Article/ Applied Article**

# Design and Performance Evaluation of Free Space Optics and Fiber Optic Communication Systems

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**Abstract: This paper focuses on the design and performance analysis of Free Space Optics (FSO) and optical fiber communication systems using the OptiSystem simulation software. The main aim of this work is to compare the performance of both communication systems under different transmission distances and attenuation conditions. In this study, both FSO and fiber optic links are designed using parameters such as bit rate and laser frequency. The performance of the systems is analysed using parameters like Q-factor, Bit Error Rate (BER), and eye diagram analysis. From the simulation results, it is observed that optical fiber communication provides stable performance with less attenuation over long distances, while the performance of FSO communication changes depending on atmospheric conditions such as fog and turbulence. The study shows that fiber optic communication is more reliable for long-distance applications, whereas FSO communication is more suitable for short-distance transmission.**

**Keywords:** Free Space Optics, Fiber Optics, BER, Q-factor, Optical Communication.

## 1. Introduction

Optical communication has become an important part of modern communication systems because it supports fast and efficient data transmission. Free Space Optics (FSO) and optical fiber communication are two commonly used technologies in this field due to their high bandwidth and better transmission speed. Different researchers have studied the behaviour of these systems under various environmental and transmission conditions. It has been observed that FSO communication is greatly affected by atmospheric conditions such as fog, haze, and turbulence, which can reduce signal quality and link reliability. In comparison, optical fiber communication provides lower attenuation and more stable performance for long-distance transmission.

In this work, both FSO and optical fiber communication systems are designed and analysed using OptiSystem simulation software. Two separate models are created for performance comparison.

Communication Systems Used

• Optical Fiber Communication System
In this system, optical signals are transmitted through an optical fiber cable.

• Free Space Optics (FSO) System
In this system, optical signals travel through free space or the atmosphere without using physical cables.

Simulation Parameters
• Data Rate – 10 Gbps
• Laser Frequency – 193.1 THz

Parameters Analysed
• Q-factor
• Bit Error Rate (BER)
• Eye Diagram Analysis

The performance of both systems is studied under different transmission distances and attenuation conditions to understand signal quality and transmission efficiency.

## 2. METHODS

**1.Fiber optic link:**

In the fiber optic link, communication takes place through an optical fiber where a Continuous Wave (CW) laser is used as the optical source. The optical signal is modulated using a Mach-Zehnder Modulator before it is transmitted through the optical fiber. The system has an attenuation constant of 0.2

dB/km. After transmission, an EDFA amplifier is used to improve the signal strength of the transmitted signal.

At the receiving end, a photodetector is used to convert the optical signal into an electrical signal. In the photodetector section, either a PIN photodiode or an Avalanche Photodiode (APD) is used. The performance of the fiber optic communication system is studied for different transmission distances such as 2 km, 5 km, 10 km, 20 km, and 30 km to observe how the signal quality and transmission performance vary with distance.

$$P(L) = P0e - \alpha L$$

Where :

$P(L)$ is the received power

$P0$ is transmitted power

$\alpha$ is attenuation coefficient

$L$ is transmission distance

Block Diagram :-

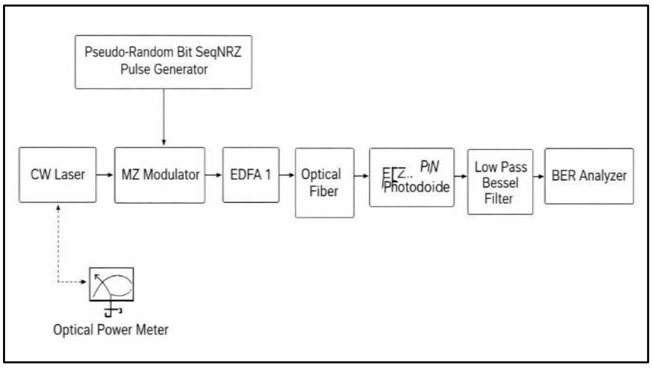


**Fig. 1:** Block diagram of Fiber Optics Communication

Simulation:-

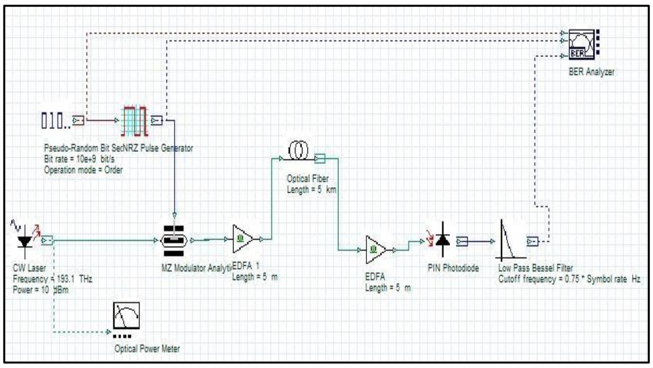


**Fig. 2:** Circuit diagram of Fiber Optics Communication

**RESULTS OF FIBER OPTICS LINK :**

Parameters:-

Bit rate = 10 Gbit/s

CW frequency = 193.1 THz

CW Laser Power = 25 dBm

Symbol rate = 10 G Symbol/sec

Extinction ratio of MZ Modulator = 30 dB

EDFA length = 5 m

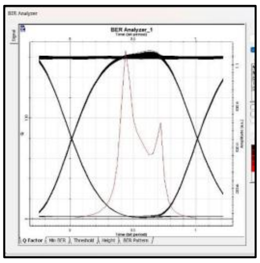

**Fig. 3:** Eye diagram for 2km

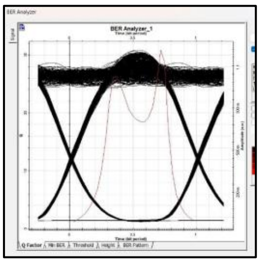

**Fig. 4:** Eye diagram for 5km

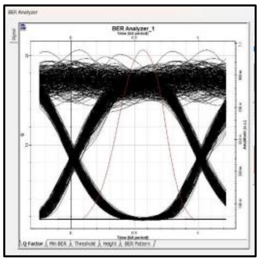

**Fig. 5:** Eye diagram for 10km

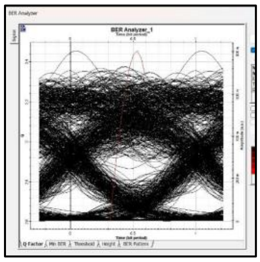

**Fig. 6:** Eye diagram for 20km

| Attenu-ation | Range (Km) | Max. Q factor | Min. BER | Eye Height | Received Power |
|---|---|---|---|---|---|
| 0.2 dB/km | 2 | 271.4 | 0 | 0.40 | 24.6 |
| | 5 | 76.91 | 0 | 0.36 | 24 |
| | 10 | 37.53 | 1.15e-30 | 0.30 | 23 |
| | 20 | 17.37 | 4.28e-68 | 0.21 | 21 |
| | 30 | 5.63 | 7.25e-09 | 0.09 | 19 |

**Table 1:** Performance Metrics Table for Attenuation 0.2dB/km

**2. FSO Link:**

In a Free Space Optics (FSO) communication system, data transmission takes place through free space, that is, without using any physical optical fiber cable. Firstly, the data generated from NRZ is modulated with a CW laser, and the optical signal is coupled and transmitted wirelessly through free space using line-of-sight communication.

To study the effects of atmospheric conditions, different attenuation values are considered such as 3 dB/km, 5 dB/km, 10 dB/km, and 15 dB/km. At the receiver end, a photodetector is used whose work is to convert the optical signal into an electrical signal. For this purpose, Avalanche Photodiode (APD) and PIN Photodiode are used.

The performance of the FSO communication system is analysed for transmission distances ranging from 1 km to 10 km under different atmospheric conditions.

$$P_r = P_t e_{-\sigma L}$$

Where:

$P_r$ is Received Power

$P_t$ is Transmitted power

$\sigma$ is Atmospheric Attenuation Coefficient

$L$ is Link Distance.

Block Diagram:-

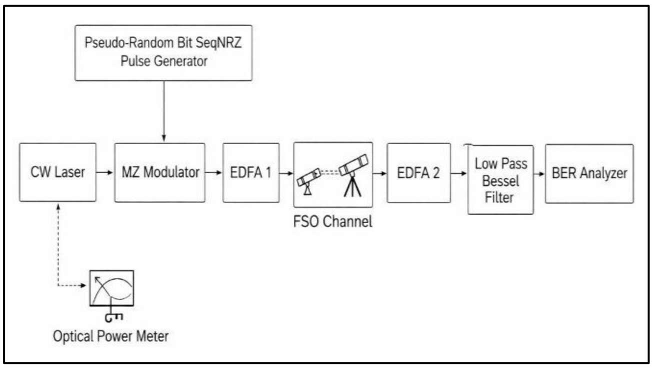


**Fig. 7:** Block diagram of Free Space Communication

Simulation:-

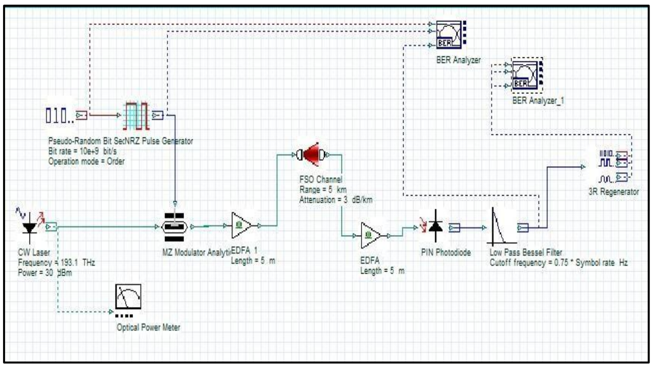


**Fig. 8:** Circuit diagram of Fiber Optics Communication

**RESULTS OF FSO LINK :**

Parameters:-
Bit rate = 10 Gbit/s
CW frequency = 193.1 THz
CW Laser Power = 25 dBm
Symbol rate = 10 G Symbol/sec
Extinction ratio of MZ Modulator = 30 dB
EDFA length = 5 m

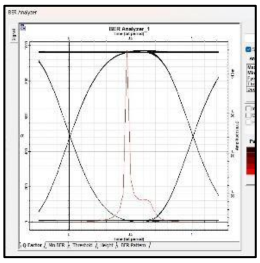

**Fig. 9:** Eye diagram for 1km

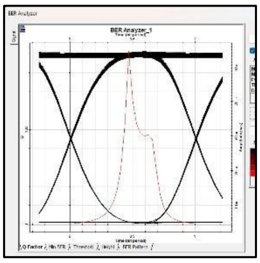

**Fig. 10:** Eye diagram for 3km

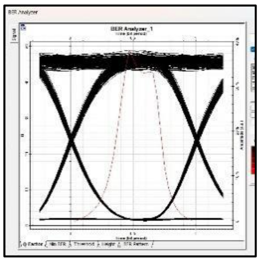

**Fig. 11:** Eye diagram for 5km

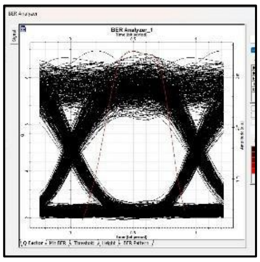

**Fig. 12:** Eye diagram for 8km

| Attenuation | Range (Km) | Max. Q factor | Min. BER | Eye Height | Received Power |
|---|---|---|---|---|---|
| 3 dB/km | 1 | 553.28 | 0 | 0.11 | 22 |
| | 3 | 90.97 | 0 | 0.06 | 16 |
| | 5 | 23.53 | 6.6e-12 | 0.01 | 10 |
| | 8 | 3.642 | 0.00 | 0.00 | 1 |

**Table 2:** Performance Metrics Table for Attenuation 3 dB/km

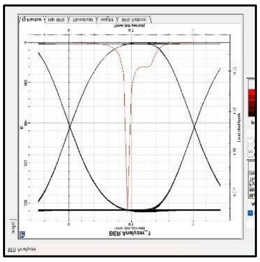

**Fig. 13:** Eye diagram for 1km

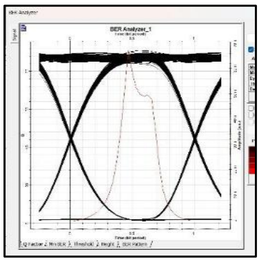

**Fig. 14:** Eye diagram for 5km

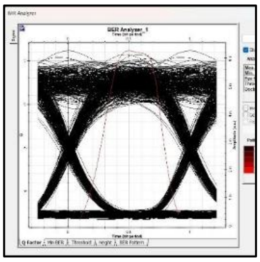

**Fig. 15:** Eye diagram for 5km

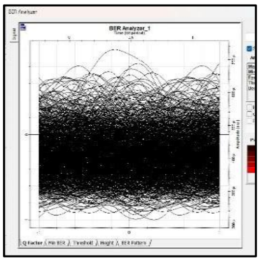

**Fig. 16:** Eye diagram for 8km

| Attenuation | Range (Km) | Max. Q factor | Min. BER | Eye Height | Received Power |
|---|---|---|---|---|---|
| 5 dB/km | 1 | 467.4 | 0 | 0.12 | 20 |
| | 3 | 44.11 | 0 | 0.02 | 10 |
| | 5 | 6.33 | 1.0266e-10 | 0.00 | 0 |
| | 8 | 0 | 1 | 0 | -15 |

**Table 3:** Performance Metrics Table for Attenuation 5dB/km

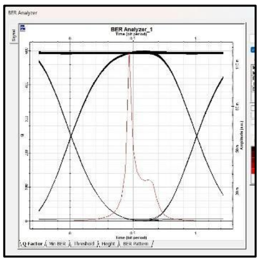

**Fig. 17:** Eye diagram for 1km

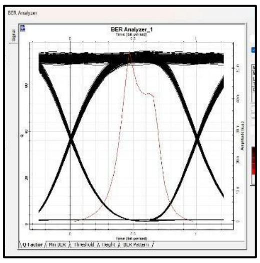

**Fig. 18:** Eye diagram for 2km

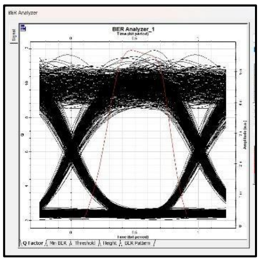

**Fig. 19:** Eye diagram for 3km

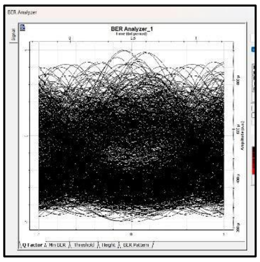

**Fig. 20:** Eye diagram for 4km

| Attenuation | Range (Km) | Max. Q factor | Min. BER | Eye Height | Received Power |
|---|---|---|---|---|---|
| 10 dB/Km | 1 | 246.36 | 0 | 0.11 | 15 |
| | 2 | 36.47 | 9.04e-292 | 0.02 | 5 |
| | 3 | 5.46 | 1.84e-08 | 0.00 | -5 |
| | 4 | 0 | 1 | 0 | -15 |

**Table 4:** Performance Metrics Table for Attenuation 10 dB/km

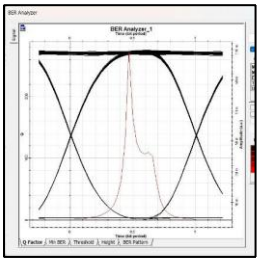

**Fig. 21:** Eye diagram for 1km

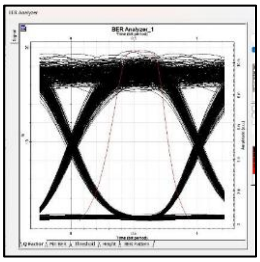

**Fig. 22:** Eye diagram for 2km

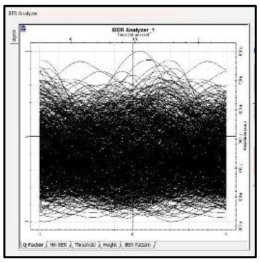

**Fig. 23:** Eye diagram for 3km

| Attenuation | Range (Km) | Max. Q factor | Min. BER | Eye Height | Received Power |
|---|---|---|---|---|---|
| 15 dB/km | 1 | 144.5 | 0 | 0.08 | 10 |
| | 2 | 9.69 | 1.24e-22 | 0 | -5 |
| | 3 | 0 | 1 | 0 | -20 |

**Table 5:** Performance Metrics Table for Attenuation 15 dB/km

## CONCULSION :

In this research paper, a detailed comparison between Free Space Optics (FSO) and Optical Fiber Communication systems is carried out using simulation analysis. From the obtained results, it is observed that FSO communication can provide high-speed data transmission without using any physical optical fiber cable, which makes the system flexible, easier to install, and cost-effective for certain applications. However, the performance of the FSO system highly depends on atmospheric conditions and attenuation factors such as fog, haze, rain, clouds, and buildings present in the communication path. These conditions affect the signal quality and reliability of communication, due to which FSO systems are mainly preferred for short-distance communication applications.

On the other hand, optical fiber communication provides better performance in terms of signal quality, reliability, and long-distance data transmission. Although it requires physical fiber cable installation, the system maintains a high Q-factor and very low BER even for larger transmission distances. Because of its stable performance and lower signal loss, optical fiber communication is more suitable for long-distance communication networks.

From the detailed analysis of both systems, it can be concluded that optical fiber communication is more suitable for long-distance and highly reliable applications, whereas FSO communication is useful for short-distance communication where wired connections are difficult to install.

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